Frenchman Gassendi. A theologian and philosopher, he was born in Provence in 1592 and died in Paris in 1655. He was a contemporary of both Galileo and Kepler, and as such he witnessed the first achievements of a revived natural science. It was about this time, after a barren interval of nearly 2,000 years, that the soil once again became fertile for the progress of scientific knowledge.

The first representatives of this new natural science, including Gassendi, revolted against the authority of Aristotle and turned to other philosophers of the classical era. Thus, Gassendi embraced the teachings of Democritus, which he at once invested with a completely materialistic form. He, too, held that the world was built of ultimate, indivisible units, or atoms, so small as to be invisible. And like Democritus, he regarded the wide variety of phenomena as the product of the variety in the arrangement and movements of atoms. The idea had already suggested itself that physical phenomena could be made intelligible in a much more concrete—one might even say, banal—way with the aid of the atomic theory. Thus, a mixture of water and wine might be compared to a mixture of two different types of sand which has been stirred so thoroughly that the two kinds of grains are completely intermingled, and distributed statistically, by pure chance. The atoms of water and wine would correspond to the grains of sand in their random and indissoluble union. Furthermore, the idea suggested itself that the states of aggregation of matter could likewise be explained by the atomic theory, even though not in the clear and intelligible manner to which we are accustomed in modern times. To-day we know that in ‘solid’ water—ice—the atoms are packed tightly in ranks and files, as it were. In ‘liquid’ water, they are also tightly packed, but are in a state of disorder, and move about in this disorderly state. Finally, in water vapour, or steam, the atoms (or more correctly, certain groups of atoms, which we call *molecules*) move in a way which may be likened to a swarm of fruit-flies, at considerable distances from each other.

This idea was taken up by other investigators, and its application to the material world progressed by leaps and bounds. For the Greeks, the conception of atoms was still the means which enabled the world, as a whole, to be understood and which accounted for observable reality. Now it became the means for the understanding of the behaviour of crude, inanimate matter.

The next scientific investigator whom we must mention was an Englishman, Robert Boyle (1627–1691). He was a chemist and physicist rather than a philosopher. His most important work concerned the theory of gases, and he discovered the law that the product of the pressure and volume of a gas at a given temperature is always constant. Chemistry is indebted to Boyle for other important discoveries, too, more especially for the introduction of the concept of the chemical elements in the modern sense. The Greeks had already associated the notion of elements with fundamental